practice today it can be difficult for most Market Participants to account for fees that are not determinable or known at the time of execution. That said, having exchange fees and rebates determinable at the time of execution will make it more likely that larger DMA brokers pass exchange fees and rebates on to their customers, including when these customers are small brokers routing their orders through them. That is because customers can better discuss fees and rebates with large DMA brokers, and the information will be more useful to customers because it is more timely. Also, while the requirements only apply to exchange fees and rebates, exchange and off-exchange trading venues compete, and transparency in exchange fees and rebates could prompt demand for greater transparency in off-exchange trading fees.[1531]

One commenter stated that rebate tiers increase aggregate liquidity, and that fee and rebate determinability will, "disrupt existing economic incentives," and, "negatively impact exchange liquidity provision and drive even more liquidity to off-exchange venues."[1532] Another commenter stated that this rule "disrupts existing economic incentives without justification,"[1533] and added that they "believe there is more aggregate liquidity in the marketplace because of the incentives provided by exchange rebate tiers." On the other hand, another commenter stated that when pricing is not determined until the end of the month, a "captive customer" is created, who "…must maintain levels of qualified trading activity or suffer an adverse economic consequence for up to an entire month's trading activity."[1534]

1531 See infra section VII.E.2 for additional discussion of the competitive effects of these amendments.

1532 See Cboe Letter II at 9-10.

1533 See Cboe Letter I at 9.

1534 BMO Letter at 4.

The Commission disagrees that fee and rebate determinability are likely to alter the economic effects related to fee and rebate tiers. Many of the incentives created by current exchange pricing schedules can be implemented by creating tier-based pricing schedules that are conditioned on historic (as opposed to future) activity, and such pricing would continue to be permissible under the amended rules.[1535] For example, a fee schedule might base the current fee or rebate tier on the share that the exchange member had of the exchange's total volume (or total consolidated volume) in the previous month.[1536] The incentives for meeting a volume tier would remain but the benefits of achieving the tier would be realized in the following month. Alternatively, a fee schedule might be based on the current month's absolute volume on the exchange (as opposed to share of volume), up until the moment of execution, which the exchange member would presumably know.[1537] Again, the incentive would be approximately the same. In general, an incentive that is based on some future quantity that cannot be known with certainty today could likely be replicated by offering the certainty equivalent,[1538] which can be calculated using a current or past quantity that can be known with certainty. This means that the uncertain portion of current fees is not strictly necessary to provide incentives. In this respect,

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1535 Such pricing can have not just benefits, but also costs. Tier-based volume pricing, for example, is used to incentivize the concentration of order flow—i.e., a member is incentivized to route orders to a particular exchange in order to qualify for a better pricing tier. This in turn creates a potential conflict of interest because the exchange member is incentivized to route customer order flow to the exchange for the purposes of tier qualification rather than maximizing other aspects of execution quality.

1536 Volume discounts like this, which are based on previous volume and then provide discounts on future purchases, have parallels in other industries (e.g., loyalty reward programs). Relative to the baseline, such a schedule would incentivize a customer to stay with an exchange for an additional month. However, there are ways exchanges might alleviate these concerns, such as a fee schedule that would induce a switch from one exchange to another.

1537 Exchanges could also create alternative incentive programs for new members provided these programs are comply with the requirements of the Exchange Act.

1538 A "certainty equivalent" is a term of art in economics, referring to the amount of a certain (that is, nonrandom) payment that must be given to an economic agent so that the agent would be indifferent between this payment and some random payoff.

any costs and benefits associated with volume-based fee tiering are not expected to change as a consequence of requiring fees and rebates to be determinable at the time of execution. Therefore, the benefits of the requirements that exchange fees and rebates be determinable at the time of execution will likely not include alleviating such "captive" customers.

One commenter stated that requiring fees to be knowable at the time of execution would make "participation in exchanges' growth programs more expensive in the initial month of participation."[1539] The Commission acknowledges that a new broker-dealer will not have a history of trading and therefore a tier schedule based on historical trading could not be implemented until the new broker-dealer has established a history; however, this history need not be long, and exchanges can cater to new and small broker-dealers by constructing tier schedules based on a short history.

One commenter stated that: "[R]equiring market participants to calculate their activity from the prior period in order to determine the volume fee and adjust their financial plans accordingly adds an unnecessary layer of complexity. The amount of effort required to understand the volume fee system, forecast volume fees for an upcoming period, and confirm that fees are indeed being calculated appropriately will especially disadvantage smaller brokers, who typically have less resources at their disposal for needless work such as this." [1540]

The Commission disagrees that the rule will add complexity. The rule removes the need for exchange members to perform forecasts in order to determine what fee they might be required to pay in a given moment. This is because, in order for a fee to be known at the time of execution as the amendments require, the fee cannot be based on activity that will happen after

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1539 See Nasdaq Letter I at 33.

1540 See Virtu Letter II at 11-12.

the execution. The Commission acknowledges, however, that fee schedules may remain complex. The rule however does not require more from small brokers than is required currently, namely it does not require them to understand the fee volume system, forecast volume fees (indeed it eliminates the need for forecasting), or to confirm that fees are being calculated appropriately. Thus the Commission does not expect this rule to disadvantage smaller brokers, and, to the extent that forecasting future market outcomes is more difficult for small brokers, this rule may make it easier for small brokers to compete.[1541]

Another commenter stated that determinable fees "would also limit exchanges' ability to incent market makers and other participants to quote at the NBBO and to do so in a large number of securities, including thinly-traded securities."[1542] The Commission disagrees because exchanges can continue to offer incentives based on past quoting at the NBBO in a large number of securities, including thinly-traded securities. Meaningful thresholds for pricing based on NBBO quoting activity can still be set based on historic activity, so that incentives to quote at the NBBO are expected to persist.[1543]

Another commenter stated that, "in order to achieve th[e] stated regulatory objective" of certainty as to an order's net fee and rebate price, and helping broker-dealers make better order routing decisions, "fees and rebates would have to be known prior to the time of execution (instead of at the point of execution, where the fee could vary based on the type of order being

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1541 See section VII.E.2.c for a discussion of the effect that fee determinability may have on competition.

1542 See Nasdaq Letter I at 33.

1543 For example, one exchange offers additional rebates to qualified market makers if, among other things, they quote at the NBBO at least 50% of the time during the month in an average of at least 2,700 symbols per day. See Nasdaq Stock Mkt. LLC, Equity 7, Sec. 114, available at https://listingcenter.nasdaq.com/rulebook/nasdaq/rules/nasdaq-equity-7 - section_114_market_quality_incentive_programs. Similar terms can be offered based on historic (rather than future) quoting.

accessed)."[1544] The Commission acknowledges that fees can vary based on order type—for example, removing hidden liquidity may incur a different fee than removing displayed liquidity, and the broker-dealer may not know whether the order will execute against hidden liquidity prior to the time of execution. In this case, there are two sources of potential uncertainty if fees are not determinable at execution: the broker-dealer's ultimate position on the exchange's fee schedule, and the type of liquidity that is accessed. Fee determinability allows broker-dealers to know, prior to execution, what their fee will be for each type of liquidity that might be accessed; the type of liquidity that is accessed may not be known until the time of execution. Resolving one source of uncertainty prior to execution (the broker-dealer's position on the exchange's fee schedule) can help broker-dealers make better routing decisions even if there remains uncertainty along other dimensions (e.g., the presence of hidden orders).

4. Acceleration and Implementation of the MDI Rules and Addition of Information About Best Odd-Lot Orders

The MDI Rules were designed to increase transparency into, among other things, the best priced quotations available in the market.[1545] The MDI Rules expanded NMS data and established a decentralized consolidation model, pursuant to which competing consolidators will eventually replace the exclusive SIPs for the collection, consolidation, and dissemination of NMS data.[1546] As discussed in section V.A, the Commission adopted a phased transition plan for

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1544 See Citadel Letter I at 25.

1545 MDI Adopting Release, supra note 10, at 18601-02, 18617; see also 17 CFR 242.600(b)(82).

1546 See MDI Adopting Release, supra note 10.

the MDI Rules,[1547] which has been delayed.[1548] Because the MDI Rules are not yet implemented, information about odd-lot orders in NMS stocks is only available on individual exchange proprietary data feeds, and market participants interested in quotation information for individual odd-lot orders must purchase these proprietary feeds.[1549] Due to the delays in the MDI Rules' implementation, as discussed in the Proposing Release,[1550] the Commission is adopting an accelerated implementation schedule, with some modifications from the proposal, so that market participants, including investors, will be provided with the enhanced transparency benefits earlier than anticipated in the MDI Rules.

The amendments will result in four changes to NMS data. Two of the changes will accelerate the implementation of specific aspects of MDI, namely the round lot definition and the inclusion of odd-lot quotations priced better than the NBBO in NMS data. This acceleration will result in realizing the economic effects of these MDI Rules sooner. The Commission acknowledges that the economic effects of the acceleration will be temporary, only lasting until the accelerated aspects of the MDI Rules would otherwise have been implemented.[1551] The amendments will also impose a new requirement on the exclusive SIPs to disseminate the accelerated odd-lot information until the exclusive SIPs are retired, the effect of which is to

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1547 See Proposing Release, supra note 11, at 80295 (describing the phased transition plan for the MDI Rules).

1548 See supra notes 74-78 and accompanying text.

1549 See MDI Adopting Release, supra note 10, at 18599. SIP data includes odd-lot transaction information but does not include odd-lot quotation information, except to the extent that odd-lot orders are aggregated into round lots pursuant to exchange rules; see also MDI Proposing Release, supra note 744, at 16739; MDI Adopting Release, supra note 10, at 18727. SIP data includes odd-lot transaction information but does not include odd-lot quotation information, except to the extent that odd-lot orders are aggregated into round lots pursuant to exchange rules; see also MDI Proposing Release, supra note 744, at 16739; MDI Adopting Release, supra note 10, at 18727.

1550 See Proposing Release, supra note 11, at 80295.

1551 See supra section V.E for further discussion on the MDI Rules Implementation.

result in the odd-lot information being disseminated sooner.[1552] The amendments, however, present the possibility that the new requirements on the SIPs can reduce competing consolidator competition if the additional requirements dissuade some market participants from choosing to become competing consolidators, which could reduce the expected benefits of the MDI Rules.[1553] The amendments will also require the dissemination of a standardized best odd-lot order or BOLO. The primary economic effect of this requirement will be to provide an additional standard benchmark that market participants could use to gauge execution quality—particularly for smaller or odd-lot orders.[1554]

Commenters questioned the sufficiency of the proposed 90-day implementation timeline for the acceleration of MDI Rules and the addition of the BOLO to NMS data. One commenter discussed the need for downstream programming changes for a variety of market participants: SIPs, recipients of market data, and broker-dealers.[1555] This commenter stated that implementation is likely to take over a year. Several other commenters stated that implementation will take longer than 90 days.[1556] One commenter discussed changes required for the SIPs: "The changes required for SIPs are relatively straightforward from a conceptual

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1552 See infra section VII.E.2.c for additional discussion of this effect. While the Rule requires the exclusive SIPs to distribute odd-lot data, the MDI Rules do not require the competing consolidators to disseminate odd-lot data. However, the MDI Adopting Release anticipated that at least one competing consolidator will do so because there would be demand for the data. See supra section VII.C.3.

1553 See infra section VII.E.2.c for additional discussion of MDI acceleration and the potential effect on competition between competing consolidators. Requiring the SIPs to disseminate odd-lot information may make the SIPs more likely to become competing consolidators and give them a first-mover advantage over other competing consolidators. However, this advantage is likely to be limited because competing consolidators can offer a lower latency than the SIPs currently provide, and can offer depth-of-book data in addition to odd-lot information.

1554 See infra section VII.D.6.a.ii for additional discussion on how the BOLO will make it easier for market centers and broker-dealers to compute statistics on price improvement relative to the best available displayed price, now required by amended Rule 605.

1555 See NYSE Letter I at 7.

1556 See FISD Letter at 3, Cboe Letter II at 11, and BlackRock Letter at 12.

perspective but will require significant undertakings before they can be implemented." This commenter then discussed the necessary steps, including: notice and lead time that data providers must give their clients; product decisions vendors need to make; development and testing for exchanges, vendors, and subscribers; traffic and capacity decisions given the change in message traffic; and communication and education for end users.[1557] Another commenter mentioned costs from converting round lots to actual share size in processing and display systems.[1558] The Operating Committee of the CTA-UTP Plans commented on the need for system design, equipment procurement, and industry testing.[1559] This commenter referenced a 2022 Odd Lots Proposal by the SIPs, which estimated a 10-12 month time frame for providing only top-of-book odd-lot quotations by the SIPs; given the additional requirements in the Proposal, the commenter estimated that implementation would take more than 12 months.

In light of these comments, the Commission is modifying the proposed compliance date for the round lot and odd-lot information definitions to extend the time for compliance. The Commission acknowledges that, all else equal, a shorter implementation timeline may result in greater total costs for the acceleration of the MDI Rules. Consistent with what commenters suggested was necessary for systems changes and testing among a variety of market participants, the adopted amendments extend the proposed compliance date for round lot definitions to approximately 12 months after the effective date of the amendments; for odd-lot information, the compliance date is extended to approximately 18 months after the effective date.[1560]

1557 See FISD Letter at 2-3.

1558 See Cboe Letter II at 11. The commenter did not provide an estimate of these costs, only stating that, "round lot conversion to actual share size will likely require considerable work by industry participants to their processing and display systems."

1559 See CTA-UTP Letter at 1.

1560 See section VI.C.

The 12-month implementation timeline for the round lot definition aligns with the compliance date for the amendments to Rule 612, allowing for the amended minimum pricing increments and new round lots to begin concurrently. This concurrence is expected to reduce the potential for operational risks and investor confusion from changing systems separately for both the amendments to Rule 612 and the acceleration of the MDI Rules, and reduces operational risks from a compressed timeline. Further, to the extent that commenters specifically discussed the timeline needed for the round lot definition implementation (as opposed to the timeline needed for both the round lot and odd-lot information definitions), no commenter stated that the round lot definition would require more than 12 months.

The longer 18-month implementation timeline for the odd-lot information definition is commensurate with the added complexity of disseminating new data fields for odd-lot quotations at multiple levels inside the NBBO. Additionally, the 18 month timeline is broadly consistent with the 10-12 month timeline that the SIPs estimated to be necessary for the 2022 Odd Lots Proposal; the longer timeline for the odd-lot information in these amendments allows for the fact that odd-lot information in these amendments includes quotations at each price level inside the NBBO, whereas the 2022 Odd Lots Proposal only included top-of-book odd-lot quotations. Further, the SIPs have been discussing the addition of odd-lot quotation information to the SIP data for several years and should be able to make the necessary adjustments to their processors in the adopted timeline. Finally, no commenter stated that the odd-lot information definition would require more than 18 months to implement. The adopted timeline allows market participants more time for each of the steps required for implementation than was initially proposed, which will address the concerns raised by commenters and help market participants to complete the tasks in a cost-effective manner. The compliance costs discussed in section VII.D.5.c reflect costs associated with the adopted timeline.

a. Round Lot Definition

As discussed in the Proposing Release,[1561] the round lot definition in the MDI Rules will result in numerous economic effects, and the amendments will result in realizing these effects sooner. The primary effects stem from the MDI Rules round lot definition shrinking the NBBO for stocks priced greater than $250.[1562] Other effects of changing the round lot definition include increased transparency and better order execution,[1563] as well as any effects from potentially having more orders routed to exchanges instead of ATSs.[1564] The costs of changing the round lot definition derive from upgrading systems to account for additional message traffic and modifying and reprogramming systems.[1565] The Commission also expects that changing the round lot definition will impact the mechanics of other rules and regulations.[1566] These economic

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1561 See Proposing Release, supra note 11, at 80330.

1562 See MDI Adopting Release, supra note 10, section V.C.1(b)(i), for the full discussion of the effect of changing the round lot size on the NBBO.

1563 See MDI Adopting Release, supra note 10, at 18744-47 for the full discussion of the effect of changing the round lot size on transparency and execution quality. See FIA PTG Letter II at 4 for agreement from a commenter.

1564 See MDI Adopting Release, supra note 10, at 18747 for the full discussion of the effect of changing the round lot size on exchange competition and order routing.

1565 See MDI Adopting Release, supra note 10, at 18748 for the full discussion of the expected costs of changing the round lot size. See also infra section VII.D.5 for an estimation and discussion of these compliance costs as they pertain to the proposed acceleration.

1566 See MDI Adopting Release, supra note 10, at 18749 for the full discussion of the effect of changing the round lot size on other rules and regulations. The round lot definition will mechanically tighten the NBBO, which is used as a reference price for numerous rules. The reference prices used for the Short Sale Circuit Breaker and LULD Plan will be affected, though these Rules will continue to function consistent with their stated purposes. The NBBO is also used as a benchmark for SRO rules such as RLPs, exchange market maker obligations, and for some order types; exchanges could propose rule changes to maintain the current operation of these rules. Finally, the round lot definition could increase the benefits of 606(b)(3) reports because it could result in an increase in the number of indications of interest in higher priced stocks that will be required to be included in 606(b)(3) reports.